PAVITHRA K M

Aspiring Data Analyst | Data Scientist | Machine Learning Enthusiast

Kuttippuram, Malappuram, Kerala, India | +91 8281256636 | pavithrakm657@gmail.com | LinkedIn: linkedin.com/in/pavithra-k-m-101016345 | GitHub: github.com/pavithrakm2004-git

PROFESSIONAL SUMMARY

Aspiring Data Analyst and Data Scientist pursuing an M.Sc. in Data Analytics and Computational Science, with a strong foundation in mathematics, machine learning, and data visualization. Proficient in Python, SQL, Power BI, and Google Cloud, with hands-on experience across the full analytics workflow — data cleaning, exploratory data analysis, predictive modeling, and dashboard development — delivered through four end-to-end classification and forecasting projects. Seeking a Data Analyst / Data Scientist / ML Engineering internship to apply analytical and problem-solving skills to real-world business and research challenges.

TECHNICAL SKILLS

- **Programming Languages:** Python, SQL
- **Data Analytics:** Data Cleaning, Exploratory Data Analysis (EDA), Statistical Analysis, Data Visualization
- **Machine Learning:** Classification, Regression, Feature Engineering, Model Evaluation, Deep Learning, Predictive Analytics
- **Tools & Platforms:** Power BI, Google Cloud Platform (BigQuery, Looker Studio), GitHub, Jupyter Notebook, Streamlit
- **Core Concepts:** Data Structures & Algorithms, Predictive Modeling, Problem Solving, Research & Analysis

EDUCATION

M.Sc. Data Analytics and Computational Science 2025 – 2027 (Expected)

Kerala University of Digital Sciences, Innovation and Technology (Digital University Kerala)

B.Sc. Mathematics 2022 – 2025

Mar Dionysius College, Pazhanji, University of Calicut

PROJECTS

Chess Piece Classification Using Machine Learning

- Built a 12-class chess piece image classifier by extracting HOG features and applying PCA for dimensionality reduction to streamline the feature space.
- Benchmarked SVM, Random Forest, and Logistic Regression classifiers to identify the best-performing model, and deployed the final pipeline as an interactive Streamlit app for real-time inference.

Handwritten Digit Recognition Using Machine Learning

- Built a digit classification system by engineering HOG and pixel-intensity feature sets, benchmarking SVM, KNN, Logistic Regression, and Random Forest algorithms to identify the most effective approach.
- Compared feature extraction strategies across multiple algorithms to inform model selection and improve classification robustness.

Heart Disease Prediction Using Patient Health Data

- Built a heart disease classification system on the Cleveland dataset (13 clinical attributes) using KNN, Logistic Regression, and Gradient Boosting, evaluated on held-out test data.
- Improved model interpretability and feature selection by applying mutual information and chi-square tests, and visualized decision boundaries, feature correlations, and comparative performance across classifiers.

Learning Unidirectional and Bidirectional Coupling of Discrete Maps (Ongoing Research)

- Researching Echo State Networks' (ESNs) ability to learn and predict coupled chaotic system dynamics across five discrete maps (Hénon, Lorenz, Mira, Nordmark, Lozi).
- Evaluating how reservoir dimensionality and topology affect prediction accuracy across unidirectionally and bidirectionally coupled systems.

CERTIFICATIONS

- Google Cloud Career Launchpad – Data Analytics Track, Google Cloud (2026)
- IBM Introduction to Git and GitHub
- NCC 'A', 'B', 'C' Certificates

LEADERSHIP & ACTIVITIES

- National Cadet Corps (NCC): Held a structured leadership role coordinating training and community engagement activities within a cadet unit, developing discipline, teamwork, and organizational skills over multiple years of service.

SOFT SKILLS

Analytical Thinking • Problem Solving • Team Collaboration • Communication • Time Management • Quick Learning

LANGUAGES

Malayalam | English